



SOLAR CELLS

Reinforcing self-assembly of hole transport molecules for stable inverted perovskite solar cells

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Power conversion efficiencies (PCEs) of inverted perovskite solar cells (PSCs) have been improved by the use of a self-assembled monolayer (SAM) hole transport layer. Long-term stability of PSCs requires keeping the SAM compact under the perovskite layer during operation. We found that strong polar solvents in the perovskite precursor desorb the SAM if it is anchored on substrates by hydrogen-bonded, rather than covalently bonded, hydroxyl groups. We used atomic-layer deposition to create an indium tin oxide substrate with a fully covalent hydroxyl-covered surface for SAM anchoring, as well as a SAM with a trimethoxysilane group that exhibited strong tridentate anchoring to the substrate. The resulting PSCs achieved PCEs of 24.8 (certified 24.6) and 23.2% with aperture areas of 0.08 and 1.01 square centimeters, respectively. The devices retained 98.9 and 98.2% of the initial PCE after 1000 hours damp-heat test and operation in maximum power point tracking at 85°C for 1200 hours under standard illumination, respectively.

One approach for improving the power conversion efficiencies (PCEs) of inverted perovskite solar cells (PSCs) has been to use self-assembled monolayers (SAMs), such as [2-(9*H*-carbazol-9-yl)ethyl]phosphonic acid (2PACz) and its derivatives, as hole transport materials (HTMs) (1, 2). The main reasons why SAMs enhance PCEs com-

pared with commonly used HTMs {such as poly[bis(4-phenyl)(2,4,6-trimethylphenyl)amine] (PTAA), nickel oxide (NiO_x), poly(3,4-ethylenedioxythiophene):polystyrene sulfonate (PEDOT:PSS)} include the following: (i) the thin HTM layer reduces the resistance of the transmission layer and thus the series resistances of the PSCs (3–5); (ii) ordered molecules induce hole

extraction by creating an interfacial dipole moment (1, 2, 5, 6); and (iii) the molecules reduce interfacial recombination with the perovskite absorption layer (5, 7, 8). The use of SAMs as HTMs has been reported in highly efficient inverted single-junction PSCs (9–12), all perovskite tandem solar cells (8, 13, 14), and perovskite-silicon tandem solar cells (5, 15–19).

However, the commercialization of highly efficient PSCs should also meet the requirement of long-term operational stability (20–22). Although studies have demonstrated that SAMs have good illumination and thermal stability themselves (23), SAM-based PSCs have not shown consistent stability compared with PSCs that use other photothermally stabilized HTMs (10, 11, 23–31). The SAMs applied in PSCs required anchoring to surface hydroxyl (OH) groups absorbed on transparent conductive metal oxides (TCO) to achieve self-assembly (2, 32–35). Thus, the stable adsorption of OH groups on the substrate is crucial to the self-assembly stability. The OH groups present on

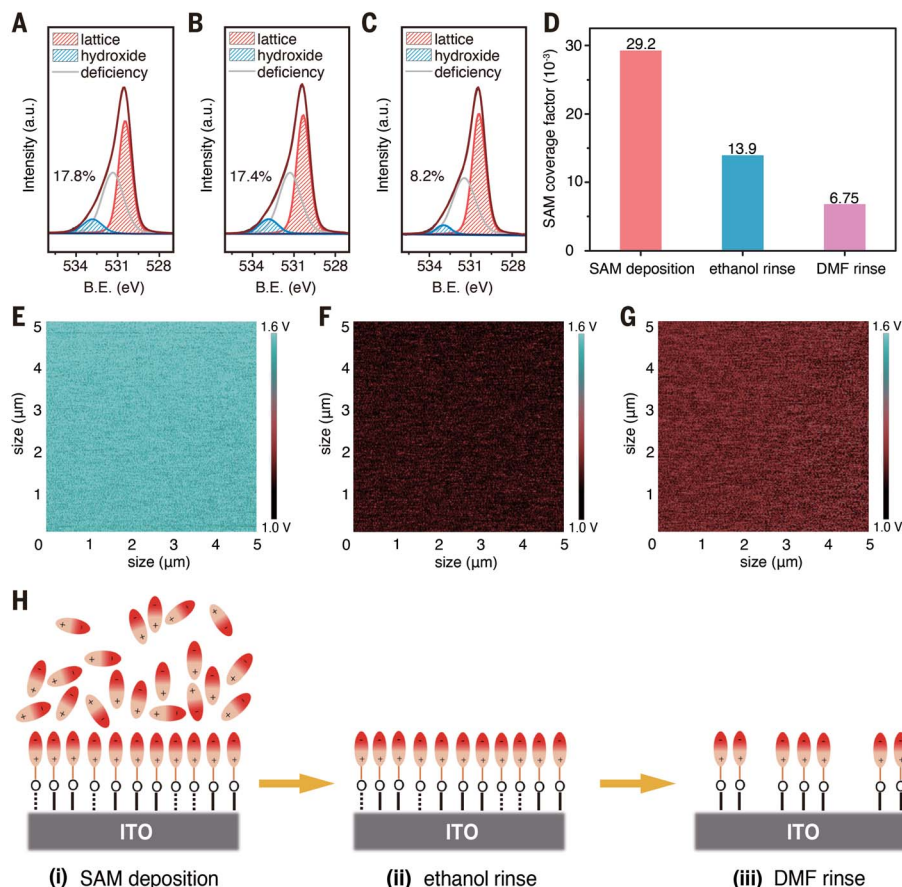
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Fig. 1. Stability studies of anchored SAM on ITO surface.

(A to C) High-resolution XPS spectra of O 1s core level of ITO, ITO after ethanol rinse, and ITO after DMF rinse, respectively. The wine, black, and navy curves are raw, fitted, and background curves, respectively. The value in the figure is the ratio of hydroxide peak area and lattice oxygen peak area. B.E., binding energy; a.u., arbitrary units. (D) The coverage factor of SAM (2PACz) on ITO and after rinsing with ethanol and DMF successively. (E to G) *xy*-plane surface potential mapping of bare ITO, ITO with 2PACz monolayer (after ethanol rinse), and after successive DMF rinse, respectively. (H) Schematic depiction of deposition and desorption of SAM on ITO surface—connected hydroxyls rinsed sequentially by ethanol then DMF. Dotted line represents a weak hydrogen bond.



a TCO surface can be either strongly bonded through chemisorption or weakly bonded through physical absorption, but the latter can be unstable in the presence of solvents (36, 37).

We found that hydrophilic OH groups weakly bonded to TCO, as well as the SAM anchored to these unstable sites, can be desorbed by strong polar perovskite solvents such as *N,N'*-dimethylformamide (DMF). Although unanchored molecules still redeposited randomly at the bottom to block electrons during perovskite crystallization, the leakage current increased as the unanchored molecules gradually desorbed from the surface, which lowered PCEs during PSC operation. We developed a metal oxide substrate with a fully covalent OH-covered surface for the fabrication of PSCs to reinforce the anchoring sites of SAM to address these issues. Moreover, we synthesized a molecule with a trimethoxysilane group, (3,6-dimethoxy-9*H*-carbazol-9-yl)trimethoxyphenylsilane (DC-TMPS), that has high binding energy with the chemisorbed OH surface through tridentate anchors, as opposed to the more commonly used SAM with phosphonic acid groups with bidentate anchors.

The fully covalent OH-covered substrate anchored by DC-TMPS ensured stable interfacial hole extraction for long-term operation and also had excellent energetic alignment with the perovskite absorption layer. The single-junction inverted PSCs had aperture areas of 0.08 and 1.01 cm² with PCEs of 24.8 (certified 24.6) and 23.2%, respectively. The stability results showed that the target PSCs retained 98.9% of the initial efficiency after damp-heat test (85°C/85% relative humidity; ISOS-D-3, where ISOS is the International Summit on Organic and Hybrid Photovoltaics Stability) for 1000 hours and retained 98.2% of the initial efficiency after operating in maximum power point tracking (MPPT) at 85°C (ISOS-L-2) for 1200 hours under AM1.5G solar light (100 mW cm⁻²).

Anchoring stability of SAM on ITO

We examined the change in adsorbed OH groups on the surface of indium tin oxide (ITO) (Fig. 1A) after rinsing with ethanol (Fig. 1B) and DMF (Fig. 1C), which are typical solvents for SAM and perovskite precursor, by measuring the peak area ratio of adsorbed OH groups and lattice O atoms (In-O and Sn-O) in O 1s core level peak in x-ray photoelectron spectroscopy (XPS) (table S1) (38, 39). After the ethanol and DMF rinses, the ratio fell from 17.8% to 17.4% and 8.2%, respectively, indicating that DMF could desorb OH from the ITO surface, but that ethanol had little impact. The results of the contact angle tests and surface energy measurements also corroborated this conclusion (fig. S1 and table S2). We demonstrated that the decreased OH groups were hydrogen-bonded species on the ITO surface, as discussed in supplementary text section 1 of the supple-

Molecule	Anchoring	Binding energy (eV)
DC-TMPS	Monodentate	-4.79
	Tridentate	-14.65
MeO-2PACz	Monodentate	-4.57
	Bidentate	-6.94

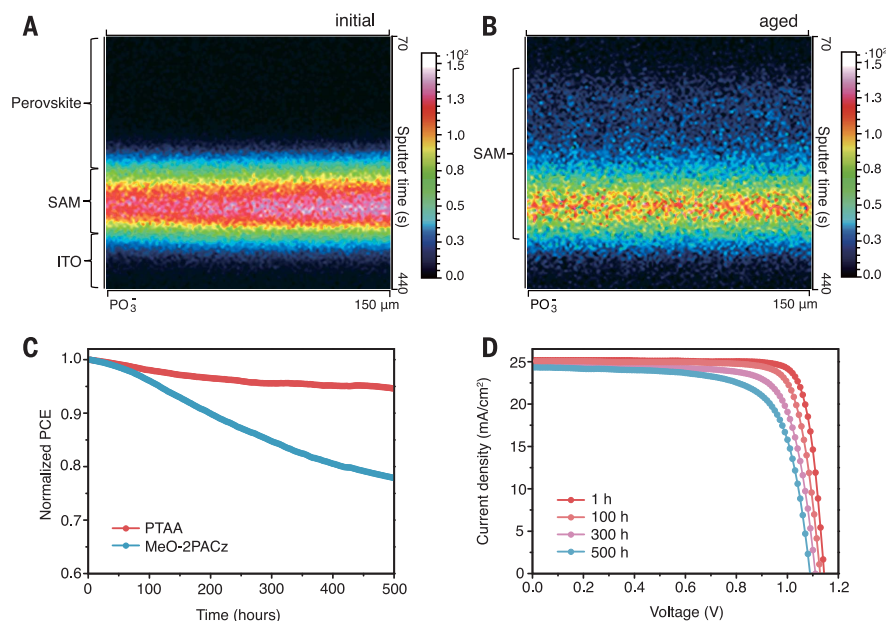


Fig. 2. Degradation mechanism of PSC with SAM HTM. (A and B) Cross-sectional TOF-SIMS mapping of SAM (PO₃⁻) in initial and aged PSC, respectively. The span of sputtering time in these images is from 70 to 440 s, which shows the distribution of MeO-2PACz molecules near the SAM layer. (C) Operational stability comparison of PSCs with PTAA HTM and SAM (MeO-2PACz) HTM. (D) Current density–voltage curves of PSC with SAM (MeO-2PACz) HTM at different aging times.

mentary materials. We used density functional theory (DFT) to calculate the binding energy of hydrogen-bonded OH on ITO, as well as the binding energy of DMF and ethanol with OH (fig. S3). The adsorption energy between OH and ITO was higher than that of ethanol and lower than that of DMF, which is consistent with DMF desorbing hydrogen-bonded OH on ITO, but ethanol having almost no effect.

We deposited a monolayer of 2PACz on ITO because its large dipole moment (1.92 D) would allow the impact of hydrogen-bonded OH to be more readily observed in contact potential difference (CPD) measurements (fig. S4 and tables S3 and S4). The CPD of ITO measured with Kelvin probe force microscopy changed from 1.56 to 1.12 V after adsorption of 2PACz (Fig. 1, E and F, and fig. S5). The coverage of 2PACz decreased, and the CPD of the ITO increased from 1.12 V to 1.23 V after additional DMF rinse of the monolayer, indicating that

some of the anchored dipolar 2PACz molecules were washed away by the solvent (Fig. 1, D and G; fig. S4; and table S3). The desorption of anchored SAM in DMF solvent was closely related to the presence of hydrogen-bonded OH. The deposition and desorption of 2PACz on the ITO surface with different solvents is schematically shown in Fig. 1H.

Photovoltaics and stability of SAM-based PSCs on commercialized ITO

Because DMF could induce desorption of some of the anchored SAM on the ITO surface, we investigated how this effect could affect fabrication of highly efficient PSCs with SAM HTMs. We deposited different thicknesses of [2-(3,6-dimethoxy-9*H*-carbazol-9-yl)ethyl]phosphonic acid (MeO-2PACz), which has been used in highly efficient inverted single-junction PSCs (9, 24, 40, 41), on ITO for fabricating PSCs. The perovskite film and solar cell characterizations are given in fig. S6. The PSCs obtained a

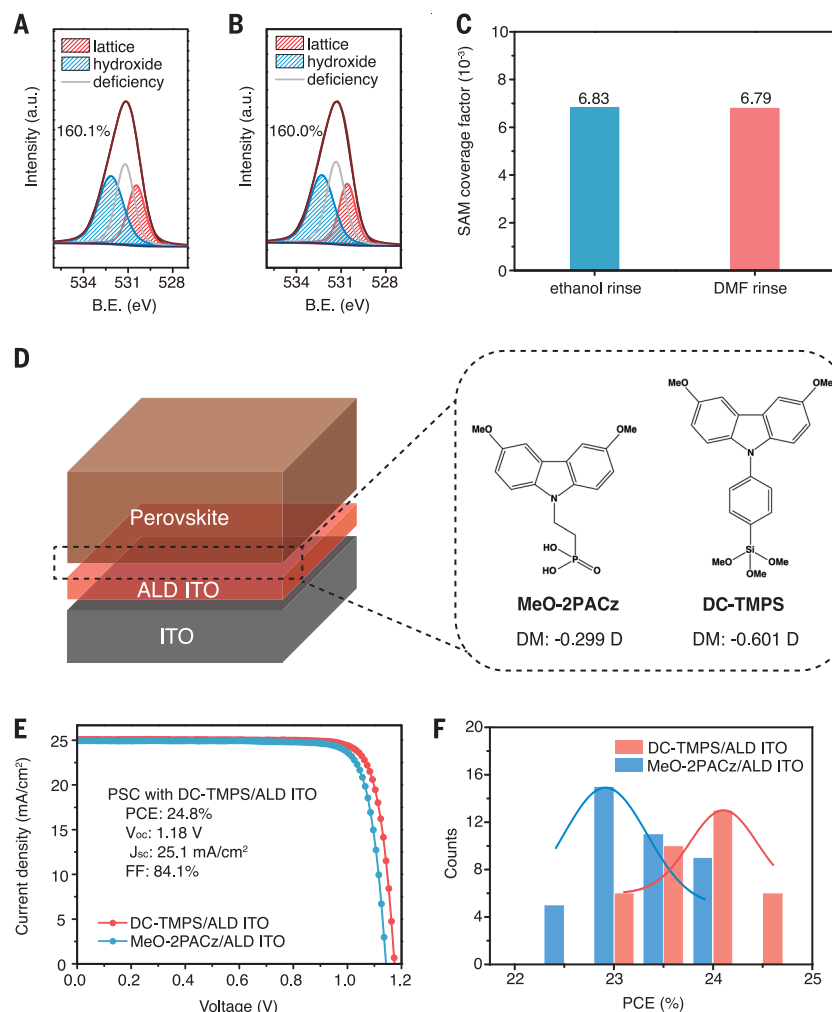


Fig. 3. Anchored SAM stability and device performance studies of ALD ITO. (A and B) High-resolution XPS spectra of O 1s core level of bare ALD ITO and ALD ITO after DMF rinse, respectively. (C) Coverage factor of SAM (2PACz) on ALD ITO after rinsing with ethanol and DMF successively. (D) Configuration schematic of perovskite/SAM/ALD ITO/ITO in PSCs using MeO-2PACz and DC-TMPS. (E) Current density-voltage curves of the best-performing PSCs with different SAM on ALD ITO. (F) Statistical distributions of the PCEs for PSCs with different SAM on ALD ITO.

maximum PCE of 23.5% with the MeO-2PACz layer without rinse treatment, whereas only 13.6% was achieved with ethanol rinse (fig. S7), indicating that highly efficient PSCs require a deposited SAM thicker than one monolayer. The vertical distribution of SAM in the PSC, characterized by time-of-flight secondary ion mass spectrometry (TOF-SIMS), showed that SAMs were mainly concentrated on the interface between perovskite absorption layer and ITO (Fig. 2A and fig. S8).

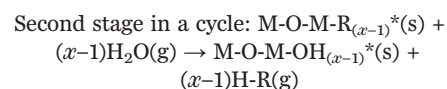
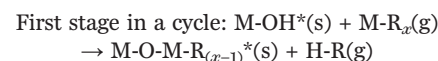
Although DMF could remove some of the SAM, the dissolved molecules were partially redeposited onto the buried interface of the perovskite layer during perovskite crystallization. To form a dense redeposited SAM layer for PSCs, the initially deposited SAM must be thicker than one monolayer, which is about 6 nm in our devices (fig. S9). However, further

increasing the thickness of MeO-2PACz would decrease PCE because of increased series resistance (fig. S10). Although highly efficient PSCs (with an ITO/MeO-2PACz/perovskite/phenethylammonium iodide (PEAI)/C₆₀/ZnO/Au structure) were obtained with thicker SAM, they exhibited severe degradation after operating in MPPT for 500 hours. Only 77.0% of the initial PCEs remained after decreases in fill factor (FF) and open-circuit voltage (V_{OC}) (Fig. 2, C and D, and table S5), where control devices (ITO/PTAA/perovskite/PEAI/C₆₀/ZnO/Au) retained 93.6% of the initial PCEs. The electroluminescence of increasingly aged PSCs with SAM HTMs gradually weakened, whereas PSCs with PTAA HTMs showed no obvious darkening, indicating that the device with SAM HTMs suffered increased nonradiative recombination during operation (fig. S11).

We exfoliated the perovskite from the substrate to compare the steady-state photoluminescence intensity of perovskite adsorption layer of PSCs containing SAM HTMs at different aging times and observed no obvious intensity changes after aging for 1 hour or 500 hours (fig. S12). These results demonstrated that the photo-generated carrier recombination in the PSCs with SAM HTMs after operating for 500 hours was not induced by degradation of electron transport materials or the perovskite absorption layer but by degradation of the SAM HTMs. We conducted TOF-SIMS to identify the vertical distribution of SAM in 500-hour-aged PSC (Fig. 2B). There was less SAM in aged PSC at the buried interface compared with the initial composition, which indicated that unanchored SAM diffused into the perovskite layer. The subsequent loss of coverage of the ITO substrate led to leakage-current increases in the devices (fig. S13), eventually resulting in V_{OC} and FF reduction.

Atomic layer deposition ITO substrate for stabilizing the anchoring of SAM

We developed ITO substrates with chemisorbed surface OH groups to address this issue. In general, it is challenging to maintain stable chemical linkages between OH groups and ITO when it is generated from inorganic metal sources through magnetron sputtering or electron beam evaporation. Organometallic sources enable the retention of covalently bonded OH groups on the surface of ITO according to the following equation



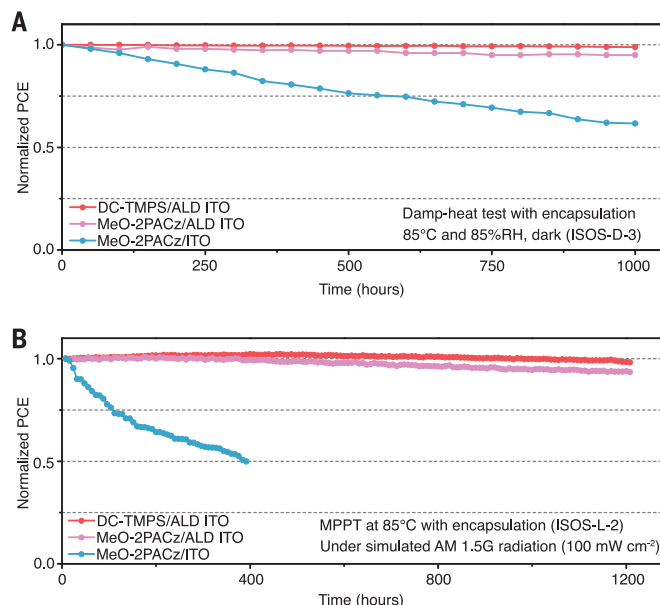
where M is a metal atom, R is an organic group, an asterisk refers to the adsorbed group, and s and g indicate solid or gas, respectively (42–44).

We used atomic layer deposition (ALD) to perform this chemical process. We prepared an additional ITO layer by ALD with an indium-to-tin deposition ratio of 9:1 (ALD ITO) on commercially available ITO. The characterizations of ALD ITO are provided in supplementary text section 2. The thickness of ALD ITO was about 9 to 10 nm. Benefiting from similar material composition with the ITO substrate, the additional thin layer of ALD ITO did not change the transmittance or resistance of ITO (fig. S15).

We probed the effect of DMF rinse on the density of OH groups on the ALD ITO surface by XPS and contact angle tests. The relative OH content changed from 160.1% to 160.0% after DMF rinse (Fig. 3, A and B, and table S6),

Fig. 4. Long-term stability studies of the PSCs.

(A) Damp-heat stability for the most-stable encapsulated PSCs using ITO with MeO-2PACz and ALD ITO with DC-TMPS and MeO-2PACz. **(B)** MPPT of the most-stable encapsulated PSCs measured at 85°C using ITO with MeO-2PACz and ALD ITO with DC-TMPS and MeO-2PACz under simulated AM1.5G illumination (100 mW cm⁻²).



and the contact angle and surface energy of ALD ITO were also nearly unchanged (fig. S19 and table S2), indicating that the surface OH groups on ALD ITO were not desorbed by DMF. We characterized the changes in SAM coverage factor and surface potential of ALD ITO anchored by the 2PACz monolayer after rinse with DMF and found that the content of 2PACz (Fig. 3C and table S7) and CPD (fig. S20) of ALD ITO was nearly unchanged. Thus, the SAM anchored with covalent OH groups on the ALD ITO surface could not be desorbed by DMF. We stored the ALD ITO substrates anchored by 2PACz in DMF solution at 85°C for 400 hours and found that the CPD of ALD ITO anchored by 2PACz increased only slightly, by 0.02 V (fig. S21).

SAM with trimethoxysilane anchoring group for strengthening the anchoring on substrate

We increased the binding energy of SAM with OH groups by designing a molecule, DC-TMPS, with a tridentate trimethoxysilane anchoring group (synthesis details and molecular characterization information are provided in supplementary text section 3). The binding energy of monodentate anchoring for DC-TMPS on ALD ITO was calculated by DFT to be -4.79 eV, compared with -4.57 eV for the commonly used bidentate SAM MeO-2PACz bearing a phosphonic acid group (fig. S25 and Table 1). During anchoring, the binding energy of DC-TMPS was calculated to be -14.65 eV, compared with -6.94 eV for MeO-2PACz.

We determined the energy levels of the perovskite absorption layer and ALD ITO anchored by different SAM with ultraviolet-visible absorption spectroscopy (fig. S26). The energy diagram of the two HTMs was obtained from

ultraviolet photoelectron spectroscopy (UPS) (fig. S27). The highest occupied molecular orbital (HOMO) level of DC-TMPS and MeO-2PACz was -5.23 and -5.15 eV, respectively, and the valence band maximum (E_{VBM}) of the perovskite measured by UPS was -5.44 eV. The work function of ALD ITO had a maximum of 5.12 eV, compared with 4.61 eV for the commercially available ITO. The low dipole moments of DC-TMPS (-0.601 D) and MeO-2PACz (-0.299 D) changed the work function of ALD ITO to 4.82 and 5.01 eV, respectively (table S4). For hole extraction from the perovskite layer, DC-TMPS exhibited a smaller potential difference than did MeO-2PACz (fig. S28).

Photovoltaics and stability of SAM-based PSCs on ALD ITO

We fabricated PSCs using ALD ITO anchored by DC-TMPS and MeO-2PACz (Fig. 3D). The maximum PCE with DC-TMPS was 24.8% for an aperture area of 0.08 cm² with a V_{OC} of 1.18 V, a short-circuit current density (J_{SC}) of 25.1 mA cm⁻², and a FF of 84.1% (Fig. 3E). For MeO-2PACz, we obtained a maximum PCE of 24.0% with a V_{OC} of 1.14 V, a J_{SC} of 25.0 mA cm⁻², and a FF of 84.0%. The ALD ITO-based PSCs using DC-TMPS have higher average values, according to the statistical distribution of the PCEs of PSCs with the two molecules (Fig. 3F). We attributed the improvement in photovoltaic performance of DC-TMPS devices to the better energetic alignment and lower interface recombination proved by quasi-Fermi level splitting analysis and capacitance-voltage test (figs. S28 to S30 and table S8) (45). We obtained a certified PCE for one device of 24.6% with a V_{OC} of 1.16 V, a J_{SC} of 25.3 mA cm⁻², and a FF of 84.0% (fig. S31). For an aperture area of 1.01 cm²,

the best-performing DC-TMPS PSC had a PCE of 23.2% with a V_{OC} of 1.17 V, a J_{SC} of 24.9 mA cm⁻², and a FF of 80.0% (fig. S32).

We evaluated the stability of our PSCs in damp-heat environment and operational conditions. For the damp-heat stability tests, all of the PSCs were encapsulated (see materials and methods in the supplementary materials for encapsulation details) and aged at 85°C under 85% relative humidity (ISOS-D-3 protocol) for 1000 hours. The PCE results showed that ALD ITO-based PSCs with DC-TMPS and MeO-2PACz HTMs retained 98.9 and 95.0% of the initial PCE respectively, whereas ITO-based PSCs with MeO-2PACz only retained 61.7% (Fig. 4A). For the operational stability test, the encapsulated PSCs were measured in MPPT under simulated AM1.5G radiation (100 mW cm⁻²) in ambient environment. The temperature of the devices was 55 ± 5°C during the long-term stability test when the room temperature was maintained at 25°C. The PSCs using ALD ITO anchored by DC-TMPS and MeO-2PACz retained 98.5 and 93.6% of the initial PCE after operating for 2000 hours, respectively, whereas ITO-based PSCs with MeO-2PACz only retained 63.2% of their initial PCE after operating for 1000 hours (fig. S33).

We further conducted the MPPT tests at elevated temperature by controlling the temperature of the devices at 85°C (ISOS-L-2). The PSCs with ALD ITO anchored by DC-TMPS and MeO-2PACz showed exceptional high-temperature durability and retained 98.2 and 93.5% of their initial efficiency after operating for 1200 hours, respectively. As controls, the ITO-based PSCs with MeO-2PACz lost 50.1% of their initial PCEs after continuous operation for 400 hours (Fig. 4B).

Discussion

We demonstrated that PCE loss of SAM-based PSCs could result from partial SAM anchoring by unstably hydrogen-bonded OH groups that were desorbed by strong polar solvents in a perovskite precursor. Although the redeposited unanchored SAM could form a dense layer at the bottom perovskite interfaces that could block electrons in fresh PSCs, its diffusion during PSC operation caused a gradual increase in leakage current. We developed an ALD ITO substrate that ensured all anchoring sites were linked with substrate by stable covalent bonds to address this issue. To further strengthen the anchoring, we created a SAM with a trimethoxysilane group with tridentate binding with the substrate. These stability improvements revealed that the ALD ITO substrate considerably improved the damp-heat and operational stability of PSCs with SAM HTMs, as the SAMs were stably anchored with the covalent hydroxyls on the surface of the substrates. Furthermore, SAM HTMs with higher binding energies with the substrates also promote stability enhancement.

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SUPPLEMENTARY MATERIALS

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Materials and Methods

Supplementary Text

Figs. S1 to S33

Tables S1 to S8

References (46–55)

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